TEEPARTHI SATYA SAI NATH

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Summary

M.Tech Computer Science student with a strong foundation in Python, SQL, data structures, algorithms, and object-oriented programming, and hands-on experience building machine learning and deep learning pipelines (TensorFlow, PyTorch, Scikit-learn) and multi-agent LLM systems with Retrieval-Augmented Generation (RAG). Experienced in data cleaning, feature engineering, exploratory data analysis, and model evaluation, with working knowledge of databases (SQL, MongoDB), Docker, and CI/CD. Strong analytical and problem-solving skills, a quick learner, and takes end-to-end ownership of projects. Seeking an internship or entry-level role in data, analytics, or AI engineering.

Education

Master of Technology in Computer Science and Engineering — Jul 2025 – Present
Amrita School of Computing, Amrita Vishwa Vidyapeetham, Bengaluru — CGPA: 8.22/10.00

Bachelor of Technology in Computer Science and Engineering — Dec 2021 – May 2025
Parul Institute of Engineering and Technology, Vadodara — CGPA: 7.47/10.00

Technical Skills

Programming Languages: Python, SQL, JavaScript (Basic)
Computer Science Fundamentals: Data Structures and Algorithms, Object-Oriented Programming (OOP), SDLC, Agile, Unit Testing and Debugging, Operating Systems and Computer Networks (Basic)
Machine Learning and AI: Machine Learning, Deep Learning, Natural Language Processing (NLP), Supervised Learning, Model Evaluation, Hyperparameter Tuning, Scikit-learn, TensorFlow, PyTorch
Generative AI and LLMs: Large Language Models (LLMs), Multi-Agent Systems, Retrieval-Augmented Generation (RAG), Model Context Protocol (MCP), LlamaIndex, Hugging Face Transformers, Prompt Engineering, Google Gemini API, Groq API
Data and Analytics: Data Cleaning, Exploratory Data Analysis (EDA), Feature Engineering, Data Preprocessing, Pandas, NumPy
Databases and DevOps: SQL, MongoDB (NoSQL), Docker, Git, GitHub Actions (CI/CD), REST API and Server-Side Design (FastMCP)
Tools: Jupyter Notebook, VS Code, Linux, Microsoft Office (Excel, PowerPoint, Word, Outlook, Teams)

Internship Experience

AI Engineer Intern — Jul 2026 – Oct 2026
Aligned Automation

- Contributed to the Bug Investigation Council, a multi-agent agentic AI pipeline that automates bug detection, root-cause investigation, and fix recommendation using coordinated LLM agents.
- Developed a domain-specific Small Language Model (SLM) for efficient, low-latency inference, covering data preparation, fine-tuning, and model evaluation.

Projects

OmniBug: Agentic Multi-Language Bug Detection Framework — 2026

- Engineered a cascading dual-agent LLM pipeline (Detector and Describer agents) that localizes, categorizes, and explains bugs across 9+ languages (Python, Java, JavaScript, TypeScript, Go, Rust, C, C++, SQL) and SmartRDI.
- Built a custom FastMCP backend API service for Retrieval-Augmented Generation (RAG), injecting domain-specific documentation into agent context to improve accuracy on niche APIs and hardware-constrained code.
- Designed a cascading model-fallback strategy with automatic escalation on rate limits, balancing throughput, reliability, and depth; classified issues into 10 structured bug categories and exported results to CSV reports for analysis.

PromptGuard: Hybrid Transformer-Based Prompt Injection Detection System — 2026

- Designed a real-time prompt injection detection framework combining a fine-tuned DistilBERT transformer with rule-based regex heuristics, trained on approximately 43K samples from six public datasets.
- Achieved a 0.93 macro F1 score with low false positives through adversarial data augmentation and threshold calibration; deployable as an API layer for LLM safety pipelines.

Certifications

- NPTEL Certified in Software Engineering